■ Microsoft Azure

Azure Data Lakehouse

Analytics Platform

Requirements Gathering Document

Project	Azure Data Lakehouse Analytics Platform
Version	v1.0
Status	Draft — For Review
Date	April 2025

01 PROJECT OVERVIEW

Project Name

Azure Data Lakehouse Analytics Platform

Objective

Build a scalable data platform using Azure services to ingest, process, and analyze data (batch + streaming), and provide actionable insights via Power BI dashboards and Machine Learning models.

02 STAKEHOLDERS

Role	Responsibility
Business Users	Consume dashboards, analyze KPIs
Data Engineers	Build & monitor ingestion pipelines
Data Scientists	Train and deploy ML models
BI Developers	Design Power BI reports & dashboards
DevOps Engineers	CI/CD, infrastructure, monitoring
Project Manager	Oversee delivery, timelines & stakeholders

03 BUSINESS REQUIREMENTS

- Support batch and real-time data ingestion
- Provide clean, reliable and governed data across all layers
- Enable interactive dashboards for business decision-making
- Support training and deployment of machine learning models

04 FUNCTIONAL REQUIREMENTS

4.1 Data Ingestion

- Ingest 24 CSV files via batch processing
- Ingest JSON unstructured data via Streaming Ingestion (Auto Loader)
- Support real-time simulation via Python Script (Spark Structured Streaming)
- Handle schema changes automatically with schema evolution

4.2 Data Storage — Azure Data Lake Gen2

- Bronze Layer — Raw data landed as-is with no transformations
- Silver Layer — Cleaned, deduplicated and validated data
- Gold Layer — Aggregated, business-ready data for reporting & ML

4.3 Data Processing — Databricks / Apache Spark

- Use Apache Spark for all ETL transformations
- Support distributed processing for large-scale workloads
- Handle both streaming and batch data within unified pipelines

4.4 Orchestration — Azure Data Factory

- Schedule and trigger pipeline executions
- Monitor job status and pipeline health
- Handle failures, retries and alerting automatically

4.5 Integration

- GitHub integration for version control of all code assets
- Databricks Repos for notebook and library synchronisation

4.6 Visualization — Power BI

- Connect directly to the Gold layer via Live Connection Stream
- Provide interactive dashboards and scheduled reports
- Support near-real-time data updates in dashboards

4.7 Machine Learning

- Use Gold layer data as the feature source for model training
- Train predictive models (e.g. Predictive Price Model)
- Deploy and serve models via Databricks ML infrastructure

05 NON-FUNCTIONAL REQUIREMENTS

Category	Requirement
Performance	Handle medium-to-large datasets; fast pipeline execution and dashboard load times
Security	Role-Based Access Control (RBAC); data encryption at rest and in transit
Scalability	Auto-scale Databricks clusters and Azure resources based on workload
Reliability	Fault-tolerant pipelines with retry logic and failure notifications
Maintainability	Centralised logging, monitoring, modular and well-documented codebase

06 DATA REQUIREMENTS

Attribute	Detail
Data Types	CSV (batch), JSON (unstructured), Streaming events
Data Volume	Medium to Large
Ingestion Frequency	Batch (scheduled) + Real-time (continuous streaming)

07 USER REQUIREMENTS

Business User

- View and interact with Power BI dashboards
- Analyze KPIs and business metrics

Data Engineer

- Build, test and deploy ingestion pipelines
- Monitor job execution and resolve failures

Data Scientist

- Access clean Gold layer data for feature engineering
- Train, evaluate and deploy ML models

08 SYSTEM REQUIREMENTS

- Azure Subscription (with sufficient quota for Databricks & Data Factory)
- Azure Data Lake Storage Gen2
- Azure Data Factory (for orchestration)
- Azure Databricks (with Apache Spark runtimes)
- Power BI Pro / Premium licence
- GitHub (organisation or personal account for Databricks Repos integration)

09 CONSTRAINTS

- Budget constraints — resource provisioning must stay within approved Azure spend limits
- Data privacy — PII and sensitive data must comply with applicable regulations
- Network performance — throughput limitations between on-premise sources and Azure

10 ASSUMPTIONS

- All required data sources are available and accessible at project start
- Azure access and necessary permissions will be provisioned prior to development
- Power BI licence is available for all BI developers and business users
- GitHub repositories will be set up and accessible for the team

11 SUCCESS CRITERIA

#	Criterion	Measurement
1	Pipelines execute successfully	Zero unhandled failures in ADF monitoring
2	Data available in all three layers	Bronze, Silver, Gold populated on schedule

3	Dashboards operational	Power BI reports refresh and display correctly
4	ML models provide insights	Models trained, deployed and returning predictions

12 FUTURE ENHANCEMENTS

- Real-time alerting system for anomaly detection and pipeline failures
- Data governance tools (e.g., Microsoft Purview) for lineage and cataloguing
- API-based ingestion to support third-party and SaaS data sources
- Advanced ML models — deep learning, AutoML and model explainability

A ARCHITECTURE DIAGRAM

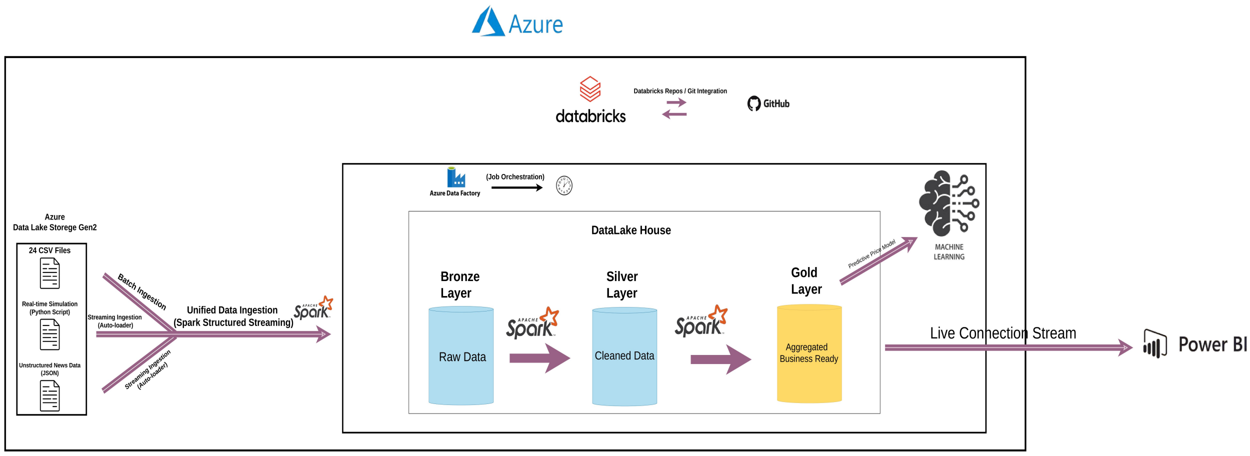

Figure 1 — Azure Data Lakehouse Analytics Platform — High-Level Architecture